Your universe at the best fit: thin constant-density shell hugging our visible sphere plus a CoE core

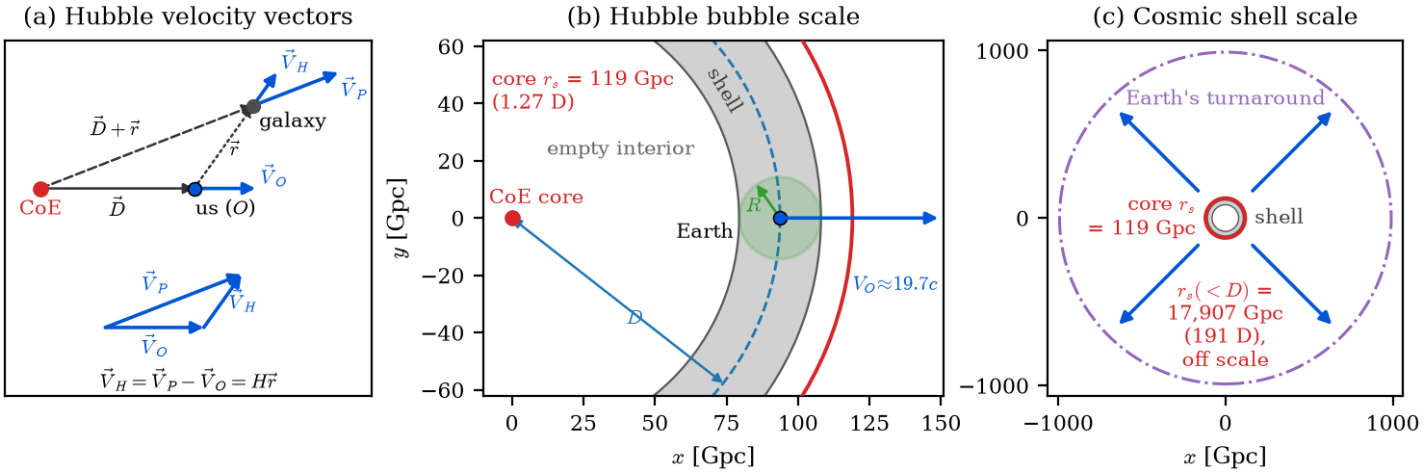


Figure 1: Three scales, left to right, at the best fit with Ω_s free; (b) and (c) are to scale in Gpc with equal axes. (a) Hubble velocity vectors. In a Sedov–Taylor blast wave (Taylor 1950; Sedov 1959) the velocity grows in proportion to distance from the centre, $\vec{V} = k \times$(position from the CoE) for some constant k. So we, at position $\vec{D}$, move with $\vec{V}_O = k\vec{D}$, and a galaxy at $\vec{D} + \vec{r}$ moves with $\vec{V}_P = k(\vec{D} + \vec{r})$. Their relative velocity is $\vec{V}_H = \vec{V}_P - \vec{V}_O = k\vec{r}$, which is Hubble's law, so $k = H$. It depends only on the separation $\vec{r}$, so the CoE drops out and every observer sees the same isotropic Hubble flow. (b) Hubble bubble: the grey shell has constant density $\Omega_s = 1.01$ (about 20 times the ordinary-matter density) from $R_{\rm in} = D - R = 79.4$ to $R_{\rm out} = D + R = 108.0$ Gpc, the interior is empty apart from the CoE core, the dashed circle of radius $D = 93.7$ Gpc runs through Earth, and the green disc is the vision sphere of radius $R = 14.3$ Gpc. The red arc is the core's own Schwarzschild radius $r_s = 2GM/c^2 = 119$ Gpc $(1.27\,D)$, already beyond Earth. Earth moves outward along $+x$, the anti-CoE direction $\ell_0 = 12.8°$ (sky directions are Galactic longitude, $\theta = \ell - \ell_0$). (c) Cosmic shell: the shell, the core's r_s and Earth's turnaround radius (991 Gpc, $10.6\,D$); the r_s of all mass inside D is 17,907 Gpc $(191\,D)$, off scale. Blue arrows are velocities in all panels. In this Newtonian model Earth moves outward at $V_O = 19.7\,c$, is bound ($(V_O/v_{\rm esc})^2 = 0.90$) and turns around after 55 ages; the mass inside has $r_s \approx 42{,}000$ Gpc, so turnaround lies inside r_s and in general relativity the region would recollapse.

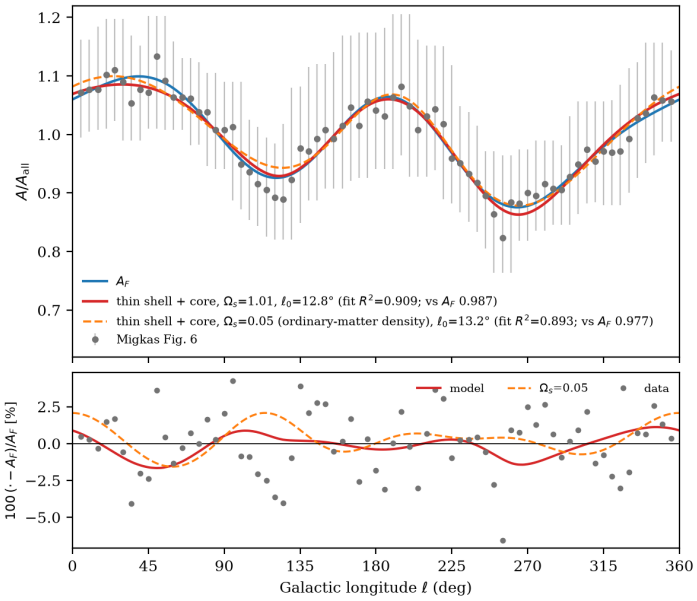


Figure 2: Migkas points with error bars, A_F, and your model A for the thin shell with a core, normalised to the sky mean. Solid: shell density free ($\Omega_s = 1.01$). Dashed: the same model with the shell density fixed at 0.05 of critical, about the density of ordinary matter. The lower panel shows data and models as percent differences from A_F. The sources spend almost all their history coasting through the empty interior, where only the CoE core pulls on them, so the core sets the pattern ($\ell_0 \approx 13°$).

Table 1: Best-fit values in your sheet's cells: integrated model (G real, from $0.01\,t_0$, fixed in space, $H_0 = 70$), shell from $D-R$ to $D+R$, ℓ_0 free. B27 is all gravitating mass inside D. The second column is the same geometry with the shell at ordinary-matter density ($\Omega_s = 0.05$).

Cell	Quantity	Ω_s free	$\Omega_s = 0.05$
C23	D/R	6.554	3.474
D23	L_c/D (core size)	0.0050 (bound)	0.0050 (bound)
B29	R	4.4127×10^{23} km (14.30 Gpc)	same
B28	D	2.8919×10^{24} km (93.72 Gpc)	1.5331×10^{24} km (49.69 Gpc)
B30	L_c	1.45×10^{22} km (0.469 Gpc)	7.67×10^{21} km (0.248 Gpc)
B27	$M(<D)$, total	3.720×10^{56} kg	4.676×10^{54} kg
		$1.87 \times 10^{26}\ M_\odot$	$2.35 \times 10^{24}\ M_\odot$
	M_c, core	2.47×10^{54} kg	2.38×10^{53} kg
	$M_c/M_{\rm crit}(D)$	0.00265	0.00172
	$M_{\rm shell}(<D)$	3.70×10^{56} kg	4.44×10^{54} kg
	$M_{\rm hole}$	5.74×10^{56} kg	2.51×10^{54} kg
BA57	Amp	−0.1240	−0.0622
BA59	φ	1.264 rad	0.933 rad
	ℓ_0	12.82°	13.25°
	Ω_s	1.012	0.05 (ordinary matter)
	ρ_s	9.32×10^{-27} kg m^{-3}	4.60×10^{-28} kg m^{-3}
	$R_{\rm in} = D - R$	79.4 Gpc	35.4 Gpc
	$R_{\rm out} = D + R$	108.0 Gpc	64.0 Gpc
	r_s, core	119 Gpc $(1.27\,D)$	11.5 Gpc $(0.23\,D)$
	r_s, all mass $< D$	17,907 Gpc $(191\,D)$	225 Gpc $(4.5\,D)$
	$(V_O/v_{\rm esc})^2$	0.901 (bound)	10.56 (unbound)
	Turnaround radius	991 Gpc $(10.6\,D)$	none
	Age t_0	13.8 Gyr	13.9 Gyr
	δ across sphere	0	0
	fit R^2 vs Migkas	0.909	0.893
	fit R^2 vs A_F	0.987	0.977

References. Taylor, G. I. 1950, Proc. R. Soc. Lond. A, 201, 159. Sedov, L. I. 1959, Similarity and Dimensional Methods in Mechanics (New York: Academic Press).